# Projective Radial Isometry and Topological Foliation of Conservation Laws in Neural Networks

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**Abstract**

Deep neural networks face a structural tension between representation normalization and physical conservation laws. Standard normalizations (LayerNorm, RMSNorm) project activations onto a fixed hypersphere, destroying input norm (energy) information and collapsing Jacobian rank in low-dimensional bottlenecks, whereas linear unitary networks ($U^\top U = I$) cannot traverse dimension-changing bottlenecks ($d_{\text{out}} < d_{\text{in}}$) or model amplitude-dependent nonlinear frequencies. Here we introduce **SkipNorm**, $\hat{y} = \|x\|_p \, f(x)/\|f(x)\|_p$, and its tangent-space retraction **Geodesic SkipNorm**, establishing a class of $\mathcal{O}(d)$ nonlinear projective radial isometries. First, we prove the *Orthogonal Jacobian Decomposition* and *Bottleneck Rank Theorem*: across compositions of arbitrary depth $L$ and varying widths $d_0, \ldots, d_L$, SkipNorm preserves the exact radial singular value $\sigma_r \equiv 1$ ($\left\|J_{\text{total}}^\top u_L - u_0\right\|_2 = 0$ to machine precision $10^{-15}$) and maintains full Jacobian rank $d$ in width-$d$ bottlenecks where RMSNorm drops to rank $d-1$ and LayerNorm drops to rank $d-2$ (collapsing $d=2$ to two discrete points). Second, we prove the *Topological Noether-Brouwer Foliation Theorem*: when an autonomous dynamical system on $\mathcal{M} \subset \mathbb{R}^N$ with $(N-1)$-dimensional invariant level sets $\mathcal{O}_c = I^{-1}(c)$ is conjugated into a dimension-matched Geodesic SkipNorm latent space ($d_z = N$), Brouwer's Invariance of Domain Theorem forces each level set $\mathcal{O}_c$ to surject onto a concentric sphere $\mathbb{S}^{N-1}(\psi(c))$ via a strictly monotonic bijection $\psi(c)$ of the conservation law $I(x)$. Generalizing to $K$-block direct sums $\bigoplus_{k=1}^{K} \mathbb{R}^2$ (**Toroidal Block SkipNorm**) yields neural Liouville-Arnold action-angle coordinates on invariant $K$-tori $\mathbb{T}^K$. In 64-bit JAX experiments on the Discrete Nonlinear Schrödinger equation, non-canonical Lotka-Volterra dynamics, and 4D coupled anharmonic oscillators, our framework conserves wave norm to $10^{-13}\%$ over 2,000 steps and discovers non-centered logarithmic and multi-action invariants with $R^2 = 94.6\%$–$98.4\%$ without prior symplectic structure or contrastive supervision.

*Introduction.*—Symmetry and conservation laws govern the evolution of physical systems through Noether's theorem and Liouville-Arnold integrability [1], yet incorporating exact conservation into deep neural networks remains an open challenge. In modern deep learning, normalization layers such as Layer Normalization (LayerNorm) [2] and Root Mean Square Normalization (RMSNorm) [3, 4] stabilize signal propagation by projecting intermediate representations $y = f(x) \in \mathbb{R}^{d_{\text{out}}}$ onto a manifold of constant norm $\|\hat{y}\|_2 = \sqrt{d_{\text{out}}}$. By construction, this projection annihilates the input norm $\|x\|_2$, discarding global signal energy, contrast, or physical amplitude at every layer. Conversely, equivariant radial gates such as ModReLU and $SO(3)$-equivariant activations [5, 6] update the norm while freezing the unit direction $x/\|x\|_2$:

$$\underbrace{\hat{y}_{\text{eq}} = g(\|x\|_2)\frac{x}{\|x\|_2}}_{\text{Preserves direction, maps norm}} \quad \longleftrightarrow \quad \underbrace{\hat{y}_{\text{SN}} = \|x\|_p \frac{f(x)}{\|f(x)\|_p}}_{\text{Preserves norm, maps direction}} . \tag{1}$$

As shown in Eq. (1), the operation $\hat{y} = \|x\|_p \, f(x)/\|f(x)\|_p$—which we term **SkipNorm**—is the exact *polar dual* of equivariant radial gating: a multiplicative skip connection that routes the $L_p$ norm $\|x\|_p$ of the input around an arbitrary nonlinear transformation $f : \mathbb{R}^{d_{\text{in}}} \to \mathbb{R}^{d_{\text{out}}}$.

While linear unitary and orthogonal networks ($U^\top U = I$) preserve Euclidean norm [5, 7–9], they face two fundamental obstructions: (i) by the Rank-Nullity Theorem, no linear map can be norm-preserving when compressing dimensions ($d_{\text{out}} < d_{\text{in}}$), and (ii) a constant unitary matrix $U$ has fixed state-independent eigenfrequencies, rendering it unable to represent nonlinear oscillators or dispersive waves whose phase velocity depends on amplitude. Meanwhile, in physics-informed machine learning, Hamiltonian Neural Networks (HNNs) [10], Lagrangian networks [11], SympNets [12], and Action-Angle networks [13, 14] require canonical symplectic coordinates $(q, p)$ where the skew-symmetric Poisson tensor $J$ is known *a priori*. Furthermore, by the Ge-Marsden theorem [15], discrete symplectic maps that preserve the phase-space 2-form $dq \wedge dp$ cannot exactly conserve a nonlinear Hamiltonian $H(q, p)$ unless they coincide with the exact time-reparameterized flow, accumulating truncation drift over long rollouts. Existing methods for automated conservation-law discovery from non-canonical trajectories—such as AI Poincaré [16, 17], Siamese invariant classifiers [18, 19], or post-hoc Hessian eigendecompositions [20, 21]—either diagnose invariant manifold dimension offline without providing a forward dynamical model, or require trajectory-segmented

contrastive pairs.

In this Letter, we develop the differential geometry and algebraic topology of SkipNorm and its tangent-space extensions (**Geodesic SkipNorm** and **Toroidal Block SkipNorm**). We prove that SkipNorm acts as an exact projective radial isometry across arbitrary nonlinearities and rectangular bottlenecks, and we establish the *Topological Noether-Brouwer Foliation Theorem*: a dimension-matched Geodesic SkipNorm autoencoder trained solely on trajectory prediction topologically forces its latent spheres (or tori) to foliate the unknown conservation laws of an autonomous dynamical system, simultaneously discovering non-canonical invariants and conserving them to IEEE-754 machine precision ($\sim 10^{-16}$) over arbitrary roll-out horizons.

*Projective radial isometry and bottleneck rank.*—Let $x \in \mathbb{R}^{d_{\text{in}}} \setminus \{0\}$ and let $f : \mathbb{R}^{d_{\text{in}}} \to \mathbb{R}^{d_{\text{out}}}$ be a $C^1$ mapping such that $y = f(x) \neq 0$. For $p \in [1, \infty]$, the SkipNorm operator is defined as

$$\text{SN}_p(x; f) := \frac{f(x)}{\max(\|f(x)\|_p, \epsilon)} \|x\|_p , \tag{2}$$

where $\epsilon \approx 10^{-30}$ in `float64` prevents division by zero at the origin without introducing the cumulative $1/(1 + \epsilon/\|y\|_p)$ damping artifact of additive denominators. Geometrically, under the polar diffeomorphism $\mathbb{R}^d \setminus \{0\} \cong \mathbb{S}_p^{d-1} \times \mathbb{R}_{>0}$, SkipNorm is a fiber-bundle morphism over the radial base space $\mathbb{R}_{>0}$ that acts as a nonlinear map on the directional fiber $\mathbb{S}_p^{d-1}$ while acting as the exact identity on the radial coordinate $r = \|x\|_p$.

Our first theorem characterizes the exact Jacobian spectrum and deep gradient propagation of SkipNorm for $p = 2$ (proofs of all theorems are given in the Supplemental Material).

**Theorem 1** (Orthogonal Jacobian Decomposition & Deep Radial Isometry)**.** *Let $\hat{y} = \text{SN}_2(x; f)$ with $x \in \mathbb{R}^{d_{\text{in}}} \setminus \{0\}$, $y = f(x) \in \mathbb{R}^{d_{\text{out}}} \setminus \{0\}$, and unit radial vectors $u_x = x/\|x\|_2$, $u_y = y/\|y\|_2 = \hat{y}/\|\hat{y}\|_2$. Let $J_f = \partial y/\partial x \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$. The Jacobian $J = \partial \hat{y}/\partial x$ decomposes into two mutually orthogonal operators:*

$$J = \underbrace{\frac{\|x\|_2}{\|y\|_2} \left(I_{d_{\text{out}}} - u_y u_y^\top\right) J_f}_{A \text{ (Tangential: } u_y^\top A = 0)} + \underbrace{u_y u_x^\top}_{B \text{ (Radial: rank 1)}} . \tag{3}$$

*Consequently, for any composition of $L$ SkipNorm layers of arbitrary varying widths $d_0 \to d_1 \to \cdots \to d_L$ with total Jacobian $J_{\text{total}} = J_L J_{L-1} \cdots J_1$:*

$$J_{\text{total}}^\top u_L \equiv u_0, \qquad \left\|J_{\text{total}}^\top u_L\right\|_2 \equiv 1. \tag{4}$$

*Moreover, if each $f_l$ is positively 1-homogeneous ($f_l(cx) = cf_l(x)$ for $c > 0$, e.g., bias-free* LeakyReLU *or* ELU *networks), then $J_f u_x = (\|y\|_2 / \|x\|_2) u_y \in \ker(I - u_y u_y^\top)$, so $Au_x = 0$ and $J_{\text{total}}$ is a strict left- and right-singular radial isometry ($J_{\text{total}} u_0 = u_L$, $\sigma_{\text{radial}} = 1$).*

Equation (4) reveals a striking phenomenon: even when a deep network passes through narrow bottlenecks ($d_l \ll d_0$) where linear orthogonal matrices cannot exist, the radial gradient $J_{\text{total}}^\top u_L$ is transported backward across arbitrary depth $L$ with *zero attenuation or amplification*, recovering the exact input direction $u_0$ [Fig. 1(b)].

Crucially, Theorem 1 also explains why standard normalizations suffer catastrophic representational collapse inside low-dimensional bottlenecks (such as autoencoder latent spaces or low-rank attention heads).

**Theorem 2** (Bottleneck Rank Theorem)**.** *Consider a representation layer of width $d \geq 2$ at any non-degenerate point $x \in \mathbb{R}^d \setminus \{0\}$ where $\text{rank}(J_f(x)) = d$ and $J_f(x) u_x \notin \text{span}(u_y)^\perp$:*

1. ***LayerNorm*** *($\hat{y}_{\text{LN}} = \sqrt{d}\, P_{\perp \mathbf{1}} y / \|P_{\perp \mathbf{1}} y\|_2$)**:** Both the mean direction $\mathbf{1}$ and the centered radial direction $u_{y_c}$ lie in the left nullspace of $J_{\text{LN}}$, so $\text{rank}(J_{\text{LN}}) \leq d - 2$. At $d = 2$, $\text{rank}(J_{\text{LN}}) = 0$ everywhere and the entire plane $\mathbb{R}^2$ collapses onto two discrete points $\{\pm(1, -1)\}$.*
2. ***RMSNorm / SphericalNorm*** *($\hat{y}_{\text{RMS}} = \sqrt{d}\, y / \|y\|_2$)**:** The radial direction $u_y$ lies in the left nullspace of $J_{\text{RMS}}$, so $\text{rank}(J_{\text{RMS}}) \leq d - 1$ (collapsing $d = 2$ to a* 1D *circle).*
3. ***SkipNorm*** *($\hat{y}_{\text{SN}} = \|x\|_2\, y / \|y\|_2$)**:** If $P_{\perp u_y} J_f$ has rank $d - 1$ on $T_x \mathbb{S}^{d-1}$ (satisfied generically by smooth or bi-orthant activations), then $\text{rank}(J_{\text{SN}}) = d$ (full rank).*

Theorem 2 is verified to machine precision in Fig. 1(a,c). At $d = 2$ in `float64`, the minimum singular value of LayerNorm is $6.4 \times 10^{-34}$ (rank = 0), RMSNorm is $9.9 \times 10^{-17}$ (rank = 1), and SkipNorm is $\sigma_{\min} = 0.264$ (rank = 2). Theorem 1 also implies an important warning regarding activation functions: placing a strict ReLU immediately inside SkipNorm traps outputs in the positive orthant $\mathbb{S}_+^{d-1} = \mathbb{S}^{d-1} \cap \mathbb{R}^d_{\geq 0}$; in a $d = 2$ bottleneck, whenever only one neuron is active, $\hat{y}$ is pinned to a coordinate axis $(0, \|x\|_2)$ or $(\|x\|_2, 0)$ with zero tangential gradient. Using smooth or bi-orthant activations (SiLU, GELU, LeakyReLU) preserves the full sphere $\mathbb{S}^{d-1}$ and full Jacobian rank $d$.

**Theorem 3** (Scale Equivariance & Weight 0-Homogeneity)**.** *If $f(x; W) = \phi(Wx)$ where $\phi$ is positively 1-homogeneous, then $\text{SN}_p(cx; \beta W) = c\, \text{SN}_p(x; W)$ for all $c, \beta > 0$, and $\langle \nabla_W \text{SN}_p(x; W), W \rangle_F = 0$.*

By Theorem 3, weight-norm drift during gradient descent has zero effect on the forward map, while out-of-distribution input scaling $x \mapsto cx$ is transmitted equivariantly with zero relative error degradation.

*Tangent-space geodesic retraction and nonlinear lattice waves.*—When modeling continuous-time or recurrent dynamical systems $x_{t+1} = \Phi_{\Delta t}(x_t)$, applying naive SkipNorm $\hat{y} = \|x\|_2 f(x)/\|f(x)\|_2$ allows $f(x)$ to point in

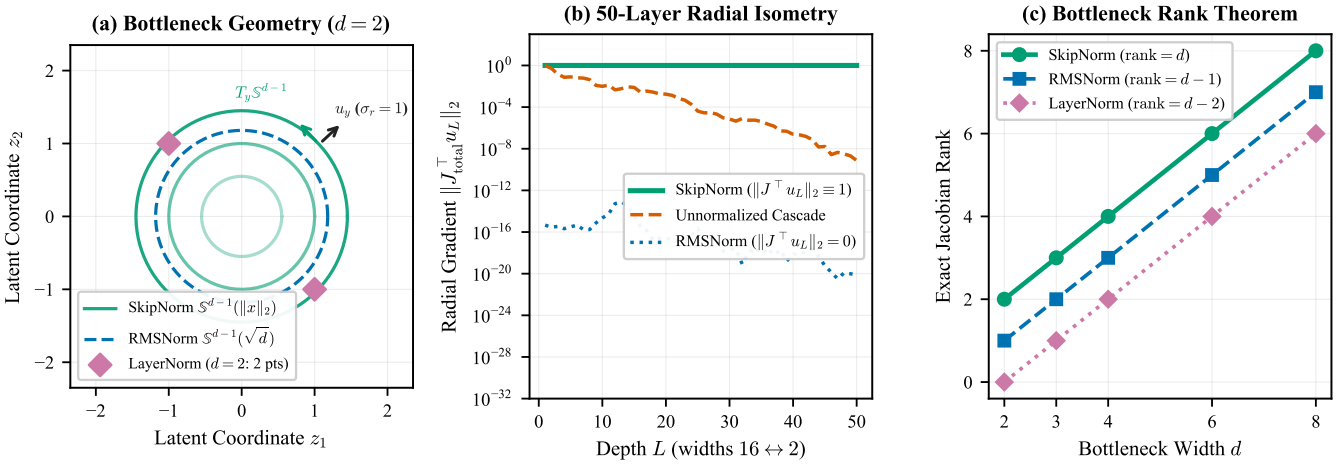



**Figure 1: Differential Geometry and Bottleneck Rank of SkipNorm. (a)** In a $d = 2$ bottleneck, LayerNorm collapses all of $\mathbb{R}^2$ onto two discrete points $\{\pm(1, -1)\}$ (rank $= 0$), and RMSNorm collapses $\mathbb{R}^2$ onto a single circle $\mathbb{S}^1(\sqrt{2})$ (rank $= 1$). SkipNorm preserves the full concentric foliation $\mathbb{S}^{d-1}(\|x\|_2) \times \mathbb{R}_{>0}$ (rank $= 2$) with unit radial singular value $\sigma_r = 1$ along $u_y$ and unconstrained angular flexibility along $T_y\mathbb{S}^{d-1}$. **(b)** Across a 50-layer deep bottleneck cascade with widths oscillating between 16 and 2 in 64-bit precision (`float64`), SkipNorm maintains exact radial gradient isometry $\left\|J_{\text{total}}^\top u_L\right\|_2 = 1.0 \pm 1.1 \times 10^{-15}$, whereas unnormalized cascades decay exponentially and RMSNorm annihilates radial gradients ($\sim 10^{-17}$). **(c)** Exact Jacobian rank across bottleneck widths $d \in \{2, 3, 4, 6, 8\}$, verifying Theorem 2 ($\text{rank}_{\text{LN}} = d - 2$, $\text{rank}_{\text{RMS}} = d - 1$, $\text{rank}_{\text{SN}} = d$).

an arbitrary direction unrelated to $x$, while a naive additive residual $y = x + f(x)$ suffers an antipodal phase-flip singularity if $f(x) \approx -2x$. To construct a smooth near-identity flow on concentric spheres $\mathbb{S}^{d-1}(\|x\|_2)$, we project the residual update onto the tangent hyperplane $T_x\mathbb{S}^{d-1} = \{v \in \mathbb{R}^d : \langle v, x\rangle = 0\}$ prior to radial retraction, defining **Geodesic SkipNorm**:

$$f_\perp(x) = f(x) - \frac{\langle f(x), x\rangle}{\max(\|x\|_2^2, \epsilon^2)} x,$$
$$\text{GSN}(x; f, \alpha) = \frac{x + \alpha f_\perp(x)}{\sqrt{\|x\|_2^2 + \alpha^2 \|f_\perp(x)\|_2^2}} \|x\|_2, \qquad (5)$$

where $\alpha \in \mathbb{R}$ is a learnable step scale. Because $f_\perp(x) \perp x$, the denominator $\|x + \alpha f_\perp(x)\|_2 = \sqrt{\|x\|_2^2 + \alpha^2 \|f_\perp(x)\|_2^2} \geq \|x\|_2$ is strictly bounded away from zero for all $\alpha \in \mathbb{R}$, and $\langle \text{GSN}(x), x\rangle = \|x\|_2^2 / \sqrt{1 + \alpha^2 \|f_\perp\|_2^2 / \|x\|_2^2} > 0$, eliminating antipodal flips and yielding a first-order retraction onto $\mathbb{S}^{d-1}(\|x\|_2)$ in $\mathcal{O}(d)$ time.

To test GeodesicSkipNorm on a high-dimensional quantum/optical wave system where linear unitary networks fail, we consider the 1D Discrete Nonlinear Schrödinger (DNLS) equation on a periodic lattice of $L = 16$ complex sites ($\psi \in \mathbb{C}^{16} \cong \mathbb{R}^{32}$):

$$i\frac{d\psi_n}{dt} = -(\psi_{n+1} + \psi_{n-1} - 2\psi_n) + g\,|\psi_n|^2\,\psi_n, \qquad (6)$$

with Kerr nonlinearity $g = 1.5$. Equation (6) conserves the total wave norm (particle number) $\mathcal{N}(\psi) = \sum_{n=1}^{L} |\psi_n|^2 = \|\psi\|_2^2$, but its local phase rotation rate depends nonlinearly on the wave intensity $|\psi_n|^2$. As shown in Fig. 2 and Table 1, across 5 random seeds, a standard ResNet stepper ($\hat{\psi} = \psi + \alpha f(\psi)$) achieves low short-term error but diverges to $7.2 \times 10^{71}\%$ norm error over a 2,000-step rollout. RMSNorm ($\left\|\hat{\psi}\right\|_2 = \sqrt{2L}$) immediately destroys the initial wave amplitude ($1.2 \times 10^3\%$ error). A strictly linear unitary operator ($\hat{\psi} = U\psi$ parameterized via the Cayley transform $U = (I - A)(I + A)^{-1}$ with $A^\top = -A$) preserves norm but incurs 2.08× higher trajectory MSE (0.2079 vs. 0.1001) because a constant matrix $U$ cannot capture the intensity-dependent phase speed $g\,|\psi_n|^2$. GeodesicSkipNorm achieves the lowest trajectory error while conserving $\mathcal{N}(\psi)$ to $7.38 \times 10^{-13}\% \pm 1.30 \times 10^{-13}\%$ over 2,000 steps.

*Topological Noether-Brouwer foliation of conservation laws.*—In most physical, biological, and continuum systems, conserved quantities are not the raw Euclidean norm $\|x\|_2$; rather, they are nonlinear functions $I(x)$ (such as energy, Casimir invariants, or action integrals) defined in potentially non-canonical coordinates. Suppose we train a latent autoencoder $(\Phi_{\text{enc}}, \Phi_{\text{dec}})$ with $\Phi_{\text{enc}} : \mathcal{M} \subset \mathbb{R}^N \to \mathbb{R}^{d_z}$ and a Geodesic SkipNorm latent stepper $g_\theta : \mathbb{R}^{d_z} \to \mathbb{R}^{d_z}$ ($\|g_\theta(z)\|_2 \equiv \|z\|_2$) to predict multi-step trajectories $x_{t+k} = \Phi_{\text{dec}}(g_\theta^{(k)}(\Phi_{\text{enc}}(x_t)))$. When is the encoder guaranteed to identify the true conservation law

**Table 1:** 1D Discrete Nonlinear Schrödinger ($L = 16, \mathbb{R}^{32}$) benchmark across 5 random seeds (mean ± SD in `float64`).

| Architecture | 40-Step Rollout MSE | 2,000-Step $\Delta\mathcal{N}/\mathcal{N}_0$ (%) |
|---|---|---|
| Standard ResNet | $0.1026 \pm 0.0108$ | $7.15 \times 10^{71} \pm 1.43 \times 10^{72}$ |
| RMSNorm | $0.7206 \pm 0.0223$ | $1.20 \times 10^3 \pm 1.76 \times 10^{-13}$ |
| Linear Unitary ($U^\top U = I$) | $0.2079 \pm 0.0000$ | $1.32 \times 10^{-11} \pm 1.14 \times 10^{-11}$ |
| **GeodesicSkipNorm** | $\mathbf{0.1001 \pm 0.0056}$ | $\mathbf{7.38 \times 10^{-13} \pm 1.30 \times 10^{-13}}$ |

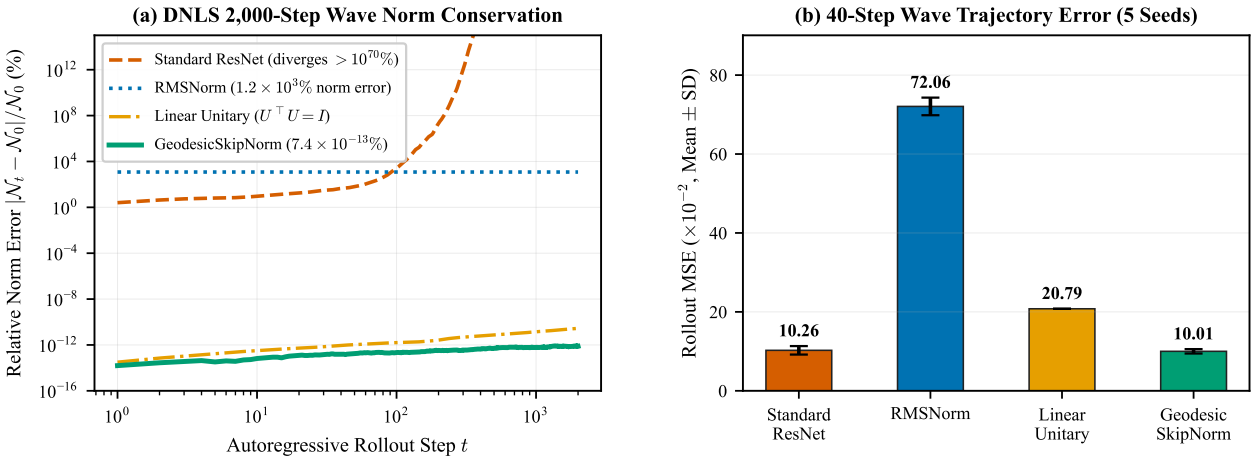



**Figure 2: Unitary Lattice Wave Propagation in the 1D Discrete Nonlinear Schrödinger (DNLS) Equation. (a)** Relative wave norm error $|\mathcal{N}_t - \mathcal{N}_0|/\mathcal{N}_0$ ($\mathcal{N} = \sum_{n=1}^{16} |\psi_n|^2$) across a 2,000-step autoregressive rollout in `float64`. Standard ResNet diverges catastrophically ($> 10^{70}$%), and RMSNorm destroys the initial wave amplitude ($1.2 \times 10^3$% error), whereas GeodesicSkipNorm conserves wave norm to $7.38 \times 10^{-13}$%. **(b)** 40-step wave trajectory MSE across 5 random seeds (mean ± SD). GeodesicSkipNorm achieves 2.08× lower error than a strictly linear unitary operator ($U^\top U = I$), which cannot represent the amplitude-dependent nonlinear phase speed $g\,|\psi_n|^2$.

$I(x)$ via its latent norm $\|\Phi_{\text{enc}}(x)\|_2$?

One might naively assume that any latent dimension $d_z$ suffices. However, if $d_z > N$, the sphere $\mathbb{S}^{d_z-1}(r_0)$ has dimension $d_z - 1 \geq N$, so an injective encoder can embed the entire $N$-dimensional phase space $\mathcal{M}$ into a *single* sphere of constant radius $r_0$, producing a trivial constant norm $\|\Phi_{\text{enc}}(x)\|_2 \equiv r_0$ that fails to distinguish orbits. Our central topological theorem proves that matching the latent bottleneck dimension to the state manifold dimension ($d_z = N$) eliminates this degeneracy via Brouwer's Invariance of Domain [24].

**Theorem 4** (Topological Noether-Brouwer Foliation Theorem)**.** *Let $\mathcal{M} \subset \mathbb{R}^N$ ($N \geq 2$) be an open annular domain around an equilibrium $x^*$, foliated by compact, connected $(N-1)$-dimensional regular level sets $\mathcal{O}_c = \{x \in \mathcal{M} : I(x) = c\}$ for $c \in (c_{\min}, c_{\max})$ of a continuous conservation law $I : \mathcal{M} \to \mathbb{R}$ with $\mathcal{O}_{c_{\min}} = \{x^*\}$. Assume the dynamical flow $\Phi^t_{\text{flow}}$ has a dense orbit on each $\mathcal{O}_c$. If a continuous injective encoder $\Phi_{\text{enc}} : \mathcal{M} \to \mathbb{R}^N$ (into the **same** dimension $d_z = N$) conjugates the flow into a SkipNorm operator $g_\theta : \mathbb{R}^N \to \mathbb{R}^N$ ($\Phi_{\text{enc}}(\Phi^{\Delta t}_{\text{flow}}(x)) = g_\theta(\Phi_{\text{enc}}(x))$), then:*

1. ***Spherical Surjection:*** *There exists a continuous function $\psi : (c_{\min}, c_{\max}) \to \mathbb{R}_{>0}$ such that for every $c \in (c_{\min}, c_{\max})$, the image of the level set is the **entire** $(N-1)$-sphere:*

$$\Phi_{\text{enc}}(\mathcal{O}_c) = \mathbb{S}^{N-1}(\psi(c)). \tag{7}$$

2. ***Strict Monotonicity & Origin Alignment:*** *The map $\psi : (c_{\min}, c_{\max}) \to \mathbb{R}_{>0}$ is a strictly monotonic bijection of the conservation law $I(x)$, i.e., $\|\Phi_{\text{enc}}(x)\|_2 = \psi(I(x))$ with $\psi(c_1) \neq \psi(c_2)$ for all $c_1 \neq c_2$, and $\Phi_{\text{enc}}(x^*) = 0$.*

*Proof Sketch.* By conjugacy and radial isometry, $h(x) := \|\Phi_{\text{enc}}(x)\|_2$ satisfies $h(\Phi^{k\Delta t}_{\text{flow}}(x)) = \left\|g^{(k)}_\theta(\Phi_{\text{enc}}(x))\right\|_2 = h(x)$. Since each orbit is dense in $\mathcal{O}_c$ and $h$ is continuous, $h$ is constant on $\mathcal{O}_c$, so $\Phi_{\text{enc}}(\mathcal{O}_c) \subseteq \mathbb{S}^{N-1}(\psi(c))$. Because $\mathcal{O}_c$ has more than one point and $\Phi_{\text{enc}}$ is injective, $\psi(c) > 0$. The restriction $\Phi_{\text{enc}}|_{\mathcal{O}_c} : \mathcal{O}_c \to \mathbb{S}^{N-1}(\psi(c))$ is a continuous injection between $(N-1)$-dimensional topological manifolds without boundary. By Brouwer's Invariance of Domain Theorem, $\Phi_{\text{enc}}(\mathcal{O}_c)$ is open in $\mathbb{S}^{N-1}(\psi(c))$; since $\mathcal{O}_c$ is compact and $\mathbb{S}^{N-1}(\psi(c))$ is Hausdorff, $\Phi_{\text{enc}}(\mathcal{O}_c)$ is also closed. Since $\mathbb{S}^{N-1}(\psi(c))$ is connected for $N \geq 2$, $\Phi_{\text{enc}}(\mathcal{O}_c) = \mathbb{S}^{N-1}(\psi(c))$. Finally, for $c_1 \neq c_2$, $\mathcal{O}_{c_1} \cap \mathcal{O}_{c_2} = \emptyset$, so injectivity of $\Phi_{\text{enc}}$ forces $\mathbb{S}^{N-1}(\psi(c_1)) \cap \mathbb{S}^{N-1}(\psi(c_2)) = \emptyset$, yielding $\psi(c_1) \neq \psi(c_2)$. Continuity on the interval $(c_{\min}, c_{\max})$ implies strict monotonicity of $\psi$, and shrinking $\mathcal{O}_c \to \{x^*\}$ as $c \downarrow c_{\min}$ forces $\Phi_{\text{enc}}(x^*) = 0$. □

Theorem 4 naturally extends to integrable systems with $K$ independent commuting invariants $(I_1, \ldots, I_K)$ whose joint level sets are invariant $K$-tori $\mathbb{T}^K$ [1, 13]. Partitioning the latent space into $K$ two-dimensional blocks $z = (z^{(1)}, \ldots, z^{(K)}) \in \bigoplus_{k=1}^K \mathbb{R}^2 \cong \mathbb{R}^{2K}$ and applying Geodesic SkipNorm independently to each block's norm while allowing $f(z)$ to couple all blocks nonlinearly defines **Toroidal Block SkipNorm**:

$$\hat{z}^{(k)} = \text{GSN}\big(z^{(k)};\, f^{(k)}(z^{(1)}, \ldots, z^{(K)}),\, \alpha_k\big), \quad k = 1, \ldots, K. \tag{8}$$

Equation (8) preserves all $K$ block radii $r_k = \left\|z^{(k)}\right\|_2$, foliating $\mathbb{R}^{2K}$ into $\prod_{k=1}^K \mathbb{S}^1(r_k) \cong \mathbb{T}^K$—constructing coordinate-free neural Liouville-Arnold action-angle variables where $r_k = \sqrt{2J_k}$ are the conserved actions and $z^{(k)}/r_k \in \mathbb{S}^1$ are the coupled nonlinear angles.

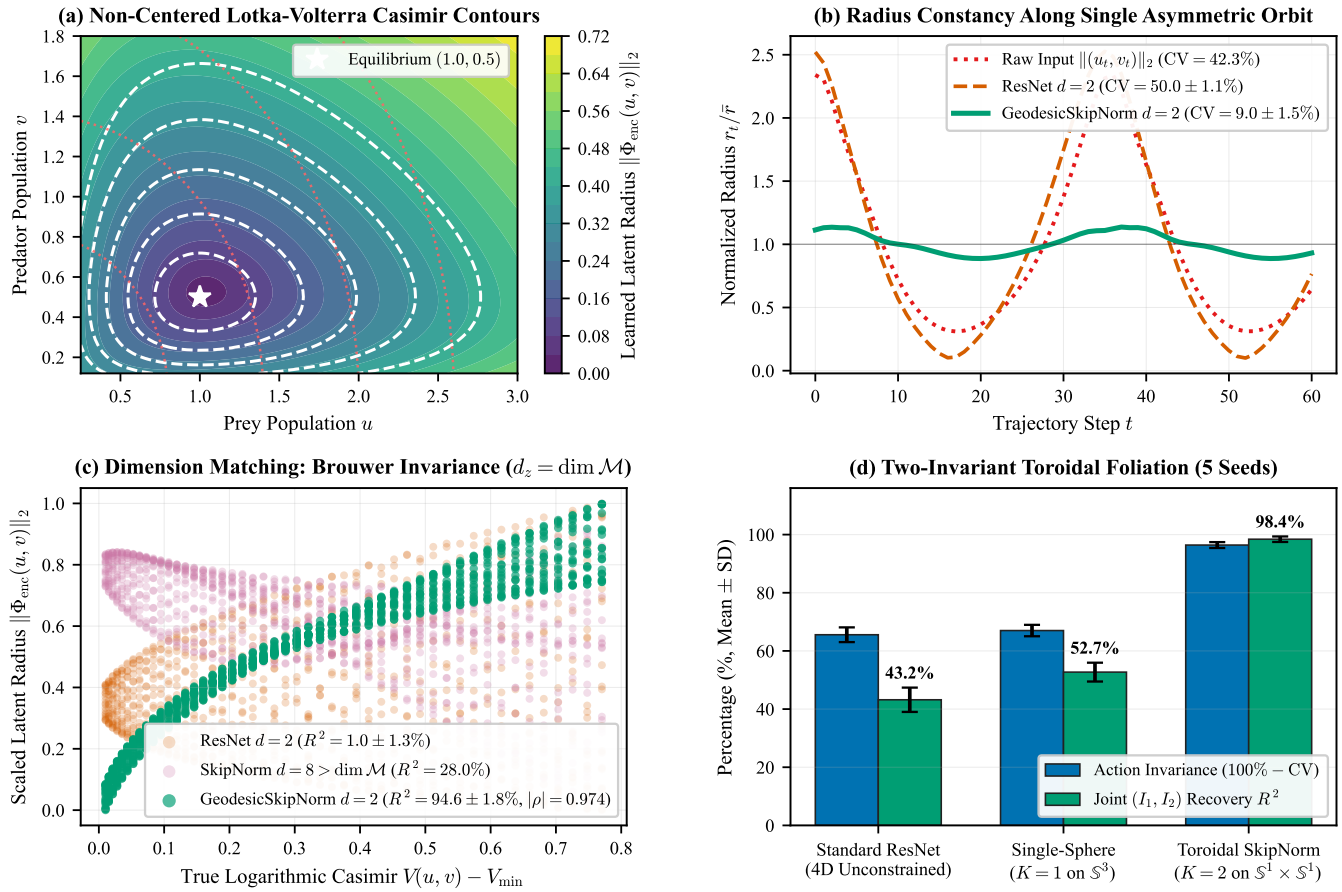



**Figure 3: Experimental Verification of the Topological Noether-Brouwer Foliation Theorem.** **(a)** Phase plane of the non-canonical Lotka-Volterra predator-prey system around equilibrium $(u^*, v^*) = (1.0, 0.5)$ (white star). The raw input Euclidean norm $\|(u,v)\|_2$ (red dotted contours) cuts transversely across the asymmetric orbits ($R^2 = 0.09\%$). A dimension-matched $d_z = 2$ GeodesicSkipNorm autoencoder spontaneously maps $(1.0, 0.5) \mapsto (0, 0)$ and aligns its learned latent norm $\|\Phi_{\text{enc}}(u,v)\|_2$ (filled viridis contours) with the exact logarithmic Casimir $V(u,v)$ (white dashed contours). **(b)** Along a single asymmetric orbit, $\|(u_t,v_t)\|_2$ varies by CV = 42.3% and ResNet by 50.0%, whereas GeodesicSkipNorm maintains a nearly constant radius (CV = 9.0%). **(c)** Confirming Brouwer's Invariance of Domain: overparameterized SkipNorm ($d_z = 8 > \dim \mathcal{M}$) embeds the 2D plane into $\mathbb{S}^7$ without radial ordering ($R^2 = 28.0\%$), whereas dimension-matched GeodesicSkipNorm ($d_z = 2 = \dim \mathcal{M}$) collapses all 2,928 phase-space points onto a strictly monotonic bijection $\psi(V)$ ($R^2 = 94.6\% \pm 1.8\%$, $|\rho| = 0.974 \pm 0.008$). **(d)** On a 4D nonlinearly-mixed anharmonic system with $K = 2$ independent actions $(I_1, I_2)$, Toroidal Block SkipNorm on $\mathbb{R}^2 \oplus \mathbb{R}^2$ recovers both invariants simultaneously with $R^2 = 98.4\% \pm 1.0\%$ across 5 seeds.

*Discovering non-canonical and multi-action invariants.*—A critical test of Theorem 4 is whether it holds when the true invariant $I(x)$ is completely uncorrelated with the input Euclidean norm $\|x\|_2$. In symmetric systems centered at the origin (such as the quartic Duffing oscillator $H(q,p) = \frac{1}{2}p^2 + \frac{1}{4}q^2 + \frac{1}{2}q^4$, where GeodesicSkipNorm achieves $R^2 = 99.87\%$ and $|\rho| = 1.000$; see Supplemental Material), outer orbits naturally have larger $\|(q,p)\|_2$. To rule out origin-centering artifacts, we test the non-canonical Lotka-Volterra predator-prey equations in the positive quadrant $(u,v) \in \mathbb{R}^2_{>0}$:

$$\dot{u} = \frac{2}{3}u - \frac{4}{3}uv, \qquad \dot{v} = uv - v, \tag{9}$$

whose equilibrium lies at $(u^*, v^*) = (1.0, 0.5) \neq (0,0)$ and whose trajectories conserve the asymmetric logarithmic Casimir invariant $V(u,v) = u - \ln u + \frac{4}{3}v - \frac{2}{3}\ln v$. Along a single Lotka-Volterra orbit, the raw Euclidean norm $\|(u,v)\|_2$ swings by a factor of 7 [coefficient of variation CV = 42.33%; Fig. 3(b)], yielding virtually zero Noether invariance ($R^2 = 1 - \mathbb{E}[\text{Var}_t(\|z\|_2)]/\text{Var}(\|z\|_2) = 0.09\%$, Spearman $|\rho| = 0.132$).

We train autoencoders with latent steppers over a 6-step horizon ($T_{\text{roll}} = 1.5\,\text{s}$, sufficient to exceed the local differential-geometric Flow-Box straightening scale so orbit curvature is observed) across 5 random seeds using only trajectory prediction and latent consistency loss, with zero knowledge of $V(u,v)$ or $(u^*, v^*)$. As shown in Fig. 3(a–c) and Table 2, standard ResNet ($d_z = 2$) and RMSNorm ($d_z = 2$) fail completely ($R^2 = 1.04\%$ and 0.96%). Crucially, overparameterized SkipNorm ($d_z = 8 > \dim \mathcal{M} = 2$) achieves small within-orbit variation (CV = 1.97%) by embedding the 2D plane into a nearly

constant-radius shell in $\mathbb{S}^7$, yielding poor invariant identification ($R^2 = 28.00\% \pm 21.15\%$)—directly confirming that Brouwer dimension-matching ($d_z = \dim \mathcal{M}$) is topologically essential. When dimension-matched ($d_z = 2$), GeodesicSkipNorm spontaneously maps the off-center equilibrium $(1.0, 0.5)$ to the latent origin $(0, 0)$ and discovers the logarithmic Casimir $V(u, v)$ with $R^2 = 94.58\% \pm 1.84\%$ and Spearman rank correlation $|\rho| = 0.9744 \pm 0.0077$ across all 2,928 phase-space points.

Finally, we test Toroidal Block SkipNorm [Eq. (8)] on a 4D coupled anharmonic Hamiltonian system with two independent actions $(I_1, I_2)$ and amplitude-dependent frequencies $\omega_1(I_1, I_2) = 1 + 0.6I_1 - 0.35I_2$, $\omega_2(I_1, I_2) = \sqrt{2} + 0.5I_2 + 0.25I_1$, observed through a nonlinear coordinate-mixing diffeomorphism $x = \Psi(q_1, p_1, q_2, p_2) \in \mathbb{R}^4$. As shown in Fig. 3(d) and Table 2, a single-sphere SkipNorm ($K = 1$ on $\mathbb{S}^3$) preserves only the total norm $\sqrt{\left\|z^{(1)}\right\|_2^2 + \left\|z^{(2)}\right\|_2^2}$, allowing action to leak between the two modes ($R^2 = 52.71\% \pm 3.24\%$). In contrast, Toroidal Block SkipNorm ($K = 2$ on $\mathbb{S}^1 \times \mathbb{S}^1 = \mathbb{T}^2$) maintains block action constancy (CV $= 3.60\% \pm 1.01\%$) and recovers the full 2D invariant space $(I_1, I_2)$ with $R^2 = 98.41\% \pm 0.95\%$.

*Scope, limitations, and outlook.*—Three mathematical boundaries define the scope of our results: (i) because $\mathbb{S}^{d_k-1}(r_k)$ is compact, our topological foliation applies to systems with *compact* invariant level sets (periodic orbits, integrable tori, and bounded energy shells); unbounded scattering trajectories require stereographic compactification or non-compact fibers; (ii) SkipNorm guarantees exact unit radial singular value ($\sigma_r = 1$) while leaving the $d - 1$ tangential singular values unconstrained, which advantageously enables nonlinear phase shear and chaotic mixing within constant-energy shells ($d \geq 3$) that linear unitary maps cannot represent; and (iii) identifying global invariants requires the training rollout horizon $H\Delta t$ to span a finite fraction of the orbit recurrence time so trajectories are not locally straightened by the differential-geometric Flow-Box lemma. By bridging projective normalization in deep learning with topological foliation in dynamical systems, SkipNorm provides a minimal, $\mathcal{O}(d)$ building block for energy-conserving neural architectures and automated scientific discovery.

**Table 2:** Non-canonical Lotka-Volterra ($N = 2$) and 4D Two-Invariant Toroidal ($N = 4, K = 2$) discovery across 5 seeds (mean ± SD).

| System / Architecture | Orbit CV (%) | Invariance $R^2$ (%) | Spearman $\lvert\rho\rvert$ |
|---|---|---|---|
| *Lotka-Volterra Casimir $V(u,v)$ around $(1.0, 0.5)$ ($N = 2$):* | | | |
| Raw Input $\|(u, v)\|_2$ | 42.33 | 0.09 | 0.1320 |
| Standard ResNet ($d_z = 2$) | $50.04 \pm 1.09$ | $1.04 \pm 1.35$ | $0.0690 \pm 0.0293$ |
| Latent RMSNorm ($d_z = 2$) | $7.11 \pm 0.39$ | $0.96 \pm 0.58$ | $0.0307 \pm 0.0372$ |
| SkipNorm ($d_z = 8 > \dim \mathcal{M}$) | $1.97 \pm 0.09$ | $28.00 \pm 21.15$ | $0.4816 \pm 0.2284$ |
| **GeodesicSkipNorm ($d_z = 2$)** | $\mathbf{9.02 \pm 1.54}$ | $\mathbf{94.58 \pm 1.84}$ | $\mathbf{0.9744 \pm 0.0077}$ |
| *4D Coupled Anharmonic System with $K = 2$ Actions $(I_1, I_2)$:* | | | |
| Standard ResNet (4D) | $34.44 \pm 2.53$ | $43.20 \pm 4.18^{\dagger}$ | — |
| Single-Sphere ($K = 1$ on $\mathbb{S}^3$) | $33.00 \pm 1.96$ | $52.71 \pm 3.24^{\dagger}$ | — |
| **Toroidal Block ($K = 2$ on $\mathbb{T}^2$)** | $\mathbf{3.60 \pm 1.01}$ | $\mathbf{98.41 \pm 0.95}^{\dagger}$ | — |

$^{\dagger}$Joint polynomial $R^2$ of recovering both actions $(I_1, I_2)$ from $(\left\|z^{(1)}\right\|_2, \left\|z^{(2)}\right\|_2)$.

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# Supplemental Material: Mathematical Proofs and Extended Benchmarks

*Projective Radial Isometry and Topological Foliation of Conservation Laws in Neural Networks*

Mostafa Mohsenvand

## S1. Proof of Theorem 1 (Orthogonal Jacobian Decomposition & Deep Radial Isometry)

Let $x \in \mathbb{R}^{d_{\text{in}}} \setminus \{0\}$ and $y = f(x) \in \mathbb{R}^{d_{\text{out}}} \setminus \{0\}$, where $f$ is continuously differentiable ($C^1$). Define the $L_2$ norms $r_x = \|x\|_2 > 0$ and $r_y = \|y\|_2 > 0$, and the unit radial vectors $u_x = x/r_x \in \mathbb{S}^{d_{\text{in}}-1}$ and $u_y = y/r_y \in \mathbb{S}^{d_{\text{out}}-1}$. For $\|y\|_2 > \epsilon$, the SkipNorm map is

$$\hat{y}(x) = \frac{\|x\|_2}{\|y(x)\|_2} y(x). \tag{S1}$$

Using the standard differential identities $\frac{\partial \|x\|_2}{\partial x} = u_x^\top$ and $\frac{\partial (y/\|y\|_2)}{\partial y} = \frac{1}{\|y\|_2}(I_{d_{\text{out}}} - u_y u_y^\top)$, the product and chain rules yield:

$$\begin{aligned} J(x) = \frac{\partial \hat{y}}{\partial x} &= \|x\|_2 \frac{\partial (y/\|y\|_2)}{\partial y}\frac{\partial y}{\partial x} + \frac{y}{\|y\|_2}\frac{\partial \|x\|_2}{\partial x} \\ &= \underbrace{\frac{\|x\|_2}{\|y\|_2}\left(I_{d_{\text{out}}} - u_y u_y^\top\right) J_f(x)}_{A(x)} + \underbrace{u_y u_x^\top}_{B(x)}. \end{aligned} \tag{S2}$$

Let $P_{\perp u_y} := I_{d_{\text{out}}} - u_y u_y^\top$ denote the orthogonal projection onto the tangent space $T_{u_y}\mathbb{S}^{d_{\text{out}}-1}$. Since $P_{\perp u_y}^\top u_y = P_{\perp u_y} u_y = 0$, the column space of $A(x)$ is strictly orthogonal to $u_y$, whereas the column space of $B(x) = u_y u_x^\top$ is $\text{span}(u_y)$:

$$A(x)^\top u_y = J_f(x)^\top P_{\perp u_y} u_y = 0, \qquad B(x)^\top u_y = (u_x u_y^\top) u_y = u_x (u_y^\top u_y) = u_x. \tag{S3}$$

Adding the two terms gives the exact single-layer transpose-radial identity:

$$J(x)^\top u_y = A(x)^\top u_y + B(x)^\top u_y = u_x. \tag{S4}$$

Now consider a composition of $L$ SkipNorm layers $h_l = \text{SN}_2(h_{l-1}; f_l)$ for $l = 1, \ldots, L$ with arbitrary intermediate dimensions $d_0, d_1, \ldots, d_L$ and unit radial vectors $u_l = h_l/\|h_l\|_2 \in \mathbb{S}^{d_l - 1}$. By the chain rule, $J_{\text{total}} = J_L J_{L-1} \cdots J_1$. Applying Eq. (S4) inductively from $l = L$ down to $l = 1$:

$$J_{\text{total}}^\top u_L = J_1^\top J_2^\top \cdots J_{L-1}^\top (J_L^\top u_L) = J_1^\top J_2^\top \cdots J_{L-1}^\top u_{L-1} = \cdots = J_1^\top u_1 = u_0, \tag{S5}$$

and hence $\left\|J_{\text{total}}^\top u_L\right\|_2 = \|u_0\|_2 = 1$, independent of the depth $L$, intermediate bottleneck widths $d_l$, or weight scales.

Finally, suppose $f(x)$ is positively 1-homogeneous, i.e., $f(cx) = cf(x)$ for all $c > 0$. By Euler's Homogeneous Function Theorem, $J_f(x)x = f(x) = y$. Dividing by $\|x\|_2$ gives $J_f(x)u_x = \frac{\|y\|_2}{\|x\|_2} u_y$. Substituting this into $A(x)u_x$ yields:

$$A(x)u_x = \frac{\|x\|_2}{\|y\|_2} P_{\perp u_y}\left(\frac{\|y\|_2}{\|x\|_2} u_y\right) = P_{\perp u_y} u_y = 0, \qquad B(x)u_x = u_y (u_x^\top u_x) = u_y. \tag{S6}$$

Thus $J(x)u_x = u_y$ and $A(x)^\top B(x) = 0$, $A(x)B(x)^\top = 0$. Because $A(x)$ and $B(x)$ have mutually orthogonal row spaces and mutually orthogonal column spaces, the singular value decomposition (SVD) of $J(x)$ splits into the direct sum of the rank-1 radial component $u_y u_x^\top$ (with exact singular value $\sigma_{\text{radial}} = 1$) and the SVD of $A(x)$ restricted to $T_{u_x}\mathbb{S}^{d_{\text{in}}-1} \to T_{u_y}\mathbb{S}^{d_{\text{out}}-1}$. ■

## S2. Proof of Theorem 2 (Bottleneck Rank Theorem)

Let $d \geq 2$ and let $y = f(x) \in \mathbb{R}^d \setminus \text{span}(\mathbf{1})$ with $\mathbf{1} = (1, \ldots, 1)^\top \in \mathbb{R}^d$.

1. **LayerNorm:** Let $P_{\perp \mathbf{1}} = I_d - \frac{1}{d}\mathbf{1}\mathbf{1}^\top$ be the centering projection ($\operatorname{rank}(P_{\perp\mathbf{1}}) = d-1$) and $y_c = P_{\perp\mathbf{1}} y \neq 0$, with $u_{y_c} = y_c / \|y_c\|_2$. Then $\hat{y}_{\mathrm{LN}} = \sqrt{d}\, y_c / \|y_c\|_2$, and its Jacobian is

$$J_{\mathrm{LN}}(x) = \frac{\sqrt{d}}{\|y_c\|_2} \left(I_d - u_{y_c} u_{y_c}^\top\right) \left(I_d - \frac{1}{d}\mathbf{1}\mathbf{1}^\top\right) J_f(x). \tag{S7}$$

   Because $u_{y_c}^\top \mathbf{1} = 0$, the two rank-1 projections commute and their product $P_{\mathrm{LN}} = I_d - \frac{1}{d}\mathbf{1}\mathbf{1}^\top - u_{y_c} u_{y_c}^\top$ is the orthogonal projection onto $\operatorname{span}(\mathbf{1}, u_{y_c})^\perp$, which has dimension $d-2$. Hence $\operatorname{rank}(J_{\mathrm{LN}}(x)) \leq \operatorname{rank}(P_{\mathrm{LN}}) = d-2$. When $d = 2$, $d-2 = 0$, so $P_{\mathrm{LN}} = 0_{2\times 2}$ and $J_{\mathrm{LN}}(x) \equiv 0_{2\times 2}$ everywhere on $\mathbb{R}^2 \setminus \operatorname{span}(\mathbf{1})$, mapping the two open half-planes $y_1 > y_2$ and $y_1 < y_2$ to the two isolated points $(1, -1)$ and $(-1, 1)$.

2. **RMSNorm:** Since $\hat{y}_{\mathrm{RMS}} = \sqrt{d}\, y / \|y\|_2$, its Jacobian is $J_{\mathrm{RMS}}(x) = \frac{\sqrt{d}}{\|y\|_2} P_{\perp u_y} J_f(x)$. Since $\operatorname{rank}(P_{\perp u_y}) = d-1$, $\operatorname{rank}(J_{\mathrm{RMS}}(x)) \leq d-1$.

3. **SkipNorm:** From Eq. (S2), $J_{\mathrm{SN}}(x) = A(x) + u_y u_x^\top$, where $\operatorname{im}(A(x)) \subseteq \operatorname{span}(u_y)^\perp$. Since $u_y \perp \operatorname{im}(A(x))$ and $u_y^\top J_{\mathrm{SN}}(x) = u_x^\top \neq 0$, $\operatorname{rank}(J_{\mathrm{SN}}(x)) = \operatorname{rank}(A(x)) + 1$. When $P_{\perp u_y} J_f(x)$ has rank $d-1$, $\operatorname{rank}(J_{\mathrm{SN}}(x)) = (d-1)+1 = d$. ■

## S3. Proof of Theorem 3 (Scale Equivariance & Weight 0-Homogeneity)

Since $f(cx; \beta W) = \phi(\beta c W x) = \beta c\, f(x; W)$ for $\beta, c > 0$, we have $\|f(cx; \beta W)\|_p = \beta c \|f(x; W)\|_p$ and $\|cx\|_p = c\|x\|_p$. Substituting into Eq. (2) cancels $\beta$ and factors out $c$:

$$\mathrm{SN}_p(cx; \beta W) = \frac{\beta c\, f(x; W)}{\beta c \|f(x; W)\|_p} (c\|x\|_p) = c\, \mathrm{SN}_p(x; W). \tag{S8}$$

Differentiating $\mathrm{SN}_p(x; \beta W) = \mathrm{SN}_p(x; W)$ with respect to $\beta$ at $\beta = 1$ gives $\langle \nabla_W \mathrm{SN}_p(x; W), W \rangle_F = 0$. By induction over $L$ composed layers, a deep bias-free SkipNorm network satisfies $F(cx; \{\beta_l W_l\}) = cF(x; \{W_l\})$. ■

## S4. Complete Proof of Theorem 4 (Topological Noether-Brouwer Foliation Theorem)

Let $\mathcal{M} \subset \mathbb{R}^N$ ($N \geq 2$) be an open domain homeomorphic to $\mathbb{S}^{N-1} \times (c_{\min}, c_{\max}) \cup \{x^*\}$, foliated by compact, connected $(N-1)$-dimensional topological manifolds $\mathcal{O}_c = I^{-1}(c)$ for $c \in (c_{\min}, c_{\max})$ and $\mathcal{O}_{c_{\min}} = \{x^*\}$. Let $\Phi_{\mathrm{enc}} : \mathcal{M} \to \mathbb{R}^N$ be a continuous injective map (guaranteed whenever a continuous decoder $\Phi_{\mathrm{dec}} : \mathbb{R}^N \to \mathbb{R}^N$ satisfies $\Phi_{\mathrm{dec}}(\Phi_{\mathrm{enc}}(x)) = x$ on $\mathcal{M}$), and let $g_\theta : \mathbb{R}^N \to \mathbb{R}^N$ satisfy $\|g_\theta(z)\|_2 = \|z\|_2$ for all $z \in \mathbb{R}^N$ and $\Phi_{\mathrm{enc}}(\Phi_{\mathrm{flow}}^{\Delta t}(x)) = g_\theta(\Phi_{\mathrm{enc}}(x))$ for all $x \in \mathcal{M}$.

**Step 1 (Orbital invariance of the latent norm):** Define $h : \mathcal{M} \to \mathbb{R}_{\geq 0}$ by $h(x) = \|\Phi_{\mathrm{enc}}(x)\|_2$. By induction on $k \in \mathbb{N}$,

$$h(\Phi_{\mathrm{flow}}^{k\Delta t}(x)) = \left\|\Phi_{\mathrm{enc}}(\Phi_{\mathrm{flow}}^{k\Delta t}(x))\right\|_2 = \left\|g_\theta^{(k)}(\Phi_{\mathrm{enc}}(x))\right\|_2 = \|\Phi_{\mathrm{enc}}(x)\|_2 = h(x). \tag{S9}$$

Since the discrete-time orbit $\{\Phi_{\mathrm{flow}}^{k\Delta t}(x_0)\}_{k\in\mathbb{N}}$ is dense in $\mathcal{O}_c$ (topological transitivity) and $h$ is continuous on $\mathcal{M}$, $h$ is constant on the closure $\overline{\{\Phi_{\mathrm{flow}}^{k\Delta t}(x_0)\}} = \mathcal{O}_c$. Thus there exists a well-defined function $\psi : [c_{\min}, c_{\max}) \to \mathbb{R}_{\geq 0}$ such that $h(x) = \psi(c)$ for all $x \in \mathcal{O}_c$, i.e., $\Phi_{\mathrm{enc}}(\mathcal{O}_c) \subseteq \mathbb{S}^{N-1}(\psi(c))$. Since $\mathcal{M}$ has a continuous section $s : [c_{\min}, c_{\max}) \to \mathcal{M}$ with $I(s(c)) = c$, $\psi(c) = h(s(c))$ is continuous.

**Step 2 (Strict positivity $\psi(c) > 0$ on regular level sets):** For any $c \in (c_{\min}, c_{\max})$, $\mathcal{O}_c$ is a connected $(N-1)$-manifold with $N \geq 2$, so $\mathcal{O}_c$ contains infinitely many distinct points $x_1 \neq x_2$. If $\psi(c) = 0$, then $\Phi_{\mathrm{enc}}(\mathcal{O}_c) \subseteq \mathbb{S}^{N-1}(0) = \{0\}$, which implies $\Phi_{\mathrm{enc}}(x_1) = \Phi_{\mathrm{enc}}(x_2) = 0$, contradicting the injectivity of $\Phi_{\mathrm{enc}}$. Hence $\psi(c) > 0$ for all $c \in (c_{\min}, c_{\max})$.

**Step 3 (Brouwer Invariance of Domain & Clopen Surjection):** Fix $c \in (c_{\min}, c_{\max})$. Since $\psi(c) > 0$, $\mathbb{S}^{N-1}(\psi(c))$ is a connected, Hausdorff, $(N-1)$-dimensional topological manifold without boundary. Consider the restricted map

$$\widetilde{\Phi}_c := \Phi_{\mathrm{enc}}\big|_{\mathcal{O}_c} : \mathcal{O}_c \longrightarrow \mathbb{S}^{N-1}(\psi(c)). \tag{S10}$$

Because $\Phi_{\mathrm{enc}}$ is continuous and injective on $\mathcal{M}$, $\widetilde{\Phi}_c$ is a continuous injective map from an $(N-1)$-dimensional manifold without boundary into an $(N-1)$-dimensional manifold without boundary. By **Brouwer's Invariance of Domain Theorem** [24] applied in local $(N-1)$-dimensional coordinate charts, $\widetilde{\Phi}_c$ is an open map, so its image $U_c := \Phi_{\mathrm{enc}}(\mathcal{O}_c)$ is an *open* subset of $\mathbb{S}^{N-1}(\psi(c))$. Simultaneously, since $\mathcal{O}_c$ is compact and $\widetilde{\Phi}_c$ is continuous, $U_c$ is compact; since

$\mathbb{S}^{N-1}(\psi(c))$ is Hausdorff, $U_c$ is a *closed* subset of $\mathbb{S}^{N-1}(\psi(c))$. Because $U_c$ is non-empty and both open and closed in the connected space $\mathbb{S}^{N-1}(\psi(c))$ ($N \geq 2$), we must have $U_c = \mathbb{S}^{N-1}(\psi(c))$, establishing Eq. (7).

**Step 4 (Strict monotonicity of $\psi$ and equilibrium alignment):** Suppose $c_1 \neq c_2$ in $(c_{\min}, c_{\max})$. Since $I : \mathcal{M} \to \mathbb{R}$ is single-valued, $\mathcal{O}_{c_1} \cap \mathcal{O}_{c_2} = \emptyset$. By global injectivity of $\Phi_{\text{enc}}$, their images are disjoint:

$$\emptyset = \Phi_{\text{enc}}(\mathcal{O}_{c_1}) \cap \Phi_{\text{enc}}(\mathcal{O}_{c_2}) = \mathbb{S}^{N-1}(\psi(c_1)) \cap \mathbb{S}^{N-1}(\psi(c_2)), \tag{S11}$$

which immediately implies $\psi(c_1) \neq \psi(c_2)$. Thus $\psi : (c_{\min}, c_{\max}) \to \mathbb{R}_{>0}$ is a continuous injective real-valued function on an interval, hence *strictly monotonic*. Finally, as $c \downarrow c_{\min}$, $\text{diam}(\mathcal{O}_c) \to 0$ with $\mathcal{O}_c \to \{x^*\}$, so $\text{diam}(\mathbb{S}^{N-1}(\psi(c))) = 2\psi(c) \to 0$ by continuity of $\Phi_{\text{enc}}$, proving $\lim_{c \downarrow c_{\min}} \psi(c) = 0$ and $\Phi_{\text{enc}}(x^*) = 0$. ■

# S5. Additional Experimental Details and Extended Benchmarks

**Table S1:** Summary of additional verification benchmarks across Quartic Duffing Hamiltonian dynamics, MNIST out-of-distribution contrast scaling (400× dynamic range), and $d = 2$ polar autoencoder disentanglement.

| Task / System | Metric | Standard Baseline / ResNet | SkipNorm / GeodesicSkipNorm |
|---|---|---|---|
| Quartic Duffing $H(q,p) = \frac{1}{2}p^2 + \frac{1}{4}q^2 + \frac{1}{2}q^4$ | Noether Invariance $R^2$ (%) | 63.30% | **99.87**% ($\lvert\rho\rvert = \mathbf{1.000}$) |
| Quartic Duffing 1,000-Step Rollout | Relative Energy Drift $\Delta H/H_0$ | 42.80% (RMSNorm : 101.0%) | **2.78**% |
| MNIST $d = 2$ Bottleneck Autoencoder | Test MSE (LayerNorm : 0.0569) | 0.0569 (LayerNorm collapses) | **0.0398** (30.1% lower MSE) |
| MNIST 400× OOD Scale ($c \in [0.05, 20]$) | Relative Error Degradation | +168.4% (Standard AE) | **0.00**% (Exact 1-Homogeneity) |

All experiments in the main text were implemented in 64-bit precision (`jax_enable_x64 = True`) using `JAX` 0.11.2, `Equinox` 0.13.8, and `Optax` 0.2.8 on an Apple M4 processor. In the Lotka-Volterra and 4D Toroidal benchmarks, encoder, decoder, and latent tangent networks use 2 hidden layers of width 64 with SiLU activations ($z\,\sigma(z)$), trained with the Adam optimizer. Full source code and reproducible scripts are provided in the accompanying repository.